

HCCM — Human Comfort Calibration Module

Parent Standard: Continuous Interaction Layer

Category: AI & Interpretation

Subcategory: Human Comfort Calibration

Type: Continuous Interaction Module

Version: 1.0

Status: Canonical · Open Module

Effective Date: 12 May 2026

Compatibility: OOF Methodology OS · Continuous Interaction Layer · CLIA · RIS · OGL · EVIP · AIL · Autonomous Runtime Systems · Realtime AI Architectures

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English

Canonical Definition

Human Comfort Calibration Module defines the structural conditions under which a continuously present intelligent system remains behaviorally compatible with human comfort, interaction pacing, interruption tolerance, cognitive load boundaries, emotional timing, and socially acceptable realtime participation.

A system satisfies HCCM only if:

- interaction pacing remains behaviorally appropriate
- interruption intensity remains bounded by human tolerance conditions
- continuous presence does not create avoidable discomfort, pressure, or overload
- proactive participation remains calibrated to human cognitive and social compatibility
- live interaction behavior remains governable not only technically, but experientially

A system that remains continuously active while disregarding human comfort conditions does not satisfy HCCM.

Module Function

HCCM defines the human comfort layer of continuous interaction governance.

It ensures that continuous interaction systems are not judged only by technical capability, low latency, or constant availability, but also by whether their live behavior remains tolerable, socially acceptable, and interaction-compatible for humans over time.

The module applies wherever a system may continuously influence:

- conversational pacing
- interruption frequency
- attention pressure
- emotional timing

- live guidance intensity
- response persistence
- ambient presence comfort
- human cognitive load during ongoing interaction

Its function is not to make a system merely pleasant.

Its function is to prevent continuous participation from becoming behaviorally excessive, socially abrasive, cognitively draining, or experientially destabilizing.

Step 1 — Define the Comfort Object

The organization must define what aspect of human interaction comfort is being governed.

Minimum requirement:

- the comfort object is explicit
- the scope of comfort calibration is structurally bounded
- undefined comfort targets are excluded from valid governance logic

The comfort object may include:

- interruption behavior
 - pacing rhythm
 - response persistence
 - timing of intervention
 - cognitive intensity
 - environmental presence behavior
 - multimodal participation behavior
 - interaction continuity over time
-

Step 2 — Define Comfort-Relevant Interaction Conditions

The system must define which interaction conditions are relevant to human comfort.

Minimum requirement:

- comfort-relevant conditions are explicit
- the system does not reduce interaction quality to technical correctness alone
- behavioral influence on the human participant remains identifiable

These conditions may include:

- interruption frequency
 - timing sensitivity
 - repetition intensity
 - proactive intervention density
 - speech pacing
-

- silence tolerance
 - escalation pressure
 - contextual emotional timing
 - continuous presence burden
-

Step 3 — Define Tolerance Boundaries

The system must define the boundaries within which live participation remains behaviorally tolerable.

Minimum requirement:

- tolerance boundaries are explicit
- behavior does not remain permanently active without calibration to human limits
- undefined behavioral intensity is excluded from valid comfort logic

Tolerance boundaries may include limits of:

- interruption rate
 - persistence of reminders
 - urgency repetition
 - attention capture intensity
 - simultaneous information load
 - response pressure
 - escalation frequency
 - presence duration without reset or relief
-

Step 4 — Define Calibration Logic

The system must define how interaction behavior is calibrated to preserve human comfort.

Minimum requirement:

- calibration logic is explicit
- the system can distinguish acceptable intensity from excessive intensity
- behavioral adaptation does not occur through arbitrary or opaque logic alone

This means the system must remain able to determine:

- when to speak
 - when to wait
 - when to interrupt
 - when to reduce intensity
 - when to maintain distance
 - when to slow interaction pacing
 - when to avoid unnecessary active presence
-
-

Step 5 — Define Cognitive Load Protection

The system must define how it avoids imposing unnecessary cognitive overload during continuous participation.

Minimum requirement:

- cognitive load conditions are explicit
- the system does not treat continuous presence as permission for unlimited engagement
- overload risk remains governable

This includes regulation of:

- information density
- multitasking pressure
- repeated prompting
- split-attention burden
- timing compression
- escalation during high-load conditions
- A continuously present system must remain aware not only of what it can do, but of what a human can comfortably sustain.

Step 6 — Preserve Comfort Traceability

The system must preserve traceability of comfort-relevant behavioral states and calibration changes.

Minimum requirement:

- comfort-related behavior remains reviewable
- changes in pacing, interruption, and intensity remain reconstructable
- later audit can determine whether live behavior remained within governed comfort boundaries

If human comfort effects cannot be reviewed, continuous interaction behavior becomes experientially opaque.

Step 7 — Restrict Invalid Comfort Design

The system must not be treated as valid if continuous presence remains technically effective but behaviorally excessive, socially abrasive, cognitively overloading, or emotionally mistimed.

Minimum requirement:

- invalid comfort conditions are identifiable
- comfort-disregarding participation is excluded
- continuous interaction without bounded human calibration is blocked, bounded, or invalidated where governance
- requires behavioral compatibility

Scenario

A realtime assistant remains continuously available during live interaction and may interrupt, remind, guide, or proactively participate over extended periods.

Application

HCCM is used to ensure:

- interruption frequency remains tolerable
 - proactive guidance does not become overbearing
 - pacing stays compatible with human attention and cognitive rhythm
 - continuous presence does not produce avoidable discomfort through excess activity
-

Result

The system gains a stronger continuous interaction architecture in which sustained participation remains socially and cognitively acceptable rather than merely fast and responsive.

Scenario

A system remains present across an environment and continuously interacts with users through multimodal cues, assistance behavior, and ongoing background participation.

Application

HCCM is used to ensure:

- presence remains behaviorally tolerable over time
 - escalation and interaction density remain bounded
 - the system does not create silent pressure through constant availability
 - comfort conditions remain reviewable as part of live interaction governance
-

Result

The environment gains a more stable human-compatible participation layer in which continuous intelligence remains supportive without becoming intrusive or exhausting.

Canonical Closing Statement

If continuous interaction does not remain calibrated to human comfort, technical presence becomes behavioral pressure rather than governed participation.

Related Documents

→ Continuous Interaction Layer (CIL™)

OOF® MIP® Protection Notice

OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

Active Protection & Compatibility Detection applies.

Canonical Source

<https://originopenfoundation.org/>